# Audio course on steroids during pregnancy. Episode four. Late preterm steroids, ALPS, diabetes, and overuse.

Companion reading handout for simultaneous listening.
Late preterm steroids are where good obstetric reasoning can become sloppy.

The early preterm story is easier. At twenty eight or thirty weeks, the respiratory benefit of a well-timed antenatal steroid course is large. In the late preterm period, from thirty four weeks through thirty six weeks and six days, the fetus is more mature. There is still respiratory morbidity to prevent, but the benefit is smaller, the exposed population is larger, and the metabolic tradeoff is more visible.

So the late preterm question is not, do steroids work?

The question is, in which patients is the respiratory benefit worth the glycemic cost and the risk of unnecessary exposure?

The central trial is ALPS.

ALPS studied singleton pregnancies from thirty four weeks to thirty six weeks and five days at high risk for late preterm delivery. The regimen was betamethasone twelve milligrams intramuscularly, with a second dose twenty four hours later.

The respiratory signal was real. The neonatal respiratory-support composite fell from fourteen point four percent to eleven point six percent. Severe respiratory complications fell from twelve point one percent to seven point nine percent.

But the glucose signal was also real. Neonatal hypoglycemia increased from fifteen percent to twenty four percent.

That is the sentence residents should remember.

Late preterm betamethasone buys respiratory benefit and spends neonatal glycemic stability.

This is not an argument against late preterm steroids. It is an argument for precision.

ALPS did not mean every patient at thirty five weeks should receive betamethasone. It enrolled selected patients at high risk of late preterm delivery. The local protocol captures this well. From thirty four weeks to thirty four weeks and six days, a first course can be given when there is a reasonable suspicion or expectation of delivery in the coming week. It is not to be given as a vague precaution. From thirty five weeks to thirty five weeks and six days, it is not routine and should be decided case by case after senior obstetric and neonatal discussion. Over thirty six weeks, the local protocol says a steroid course will not be given.

That local rule is important because it makes the resident ask the right question: is delivery likely soon, or are we just uncomfortable?

Late preterm steroid overuse happens when discomfort becomes an indication.

Now add diabetes.

Steroids increase maternal glucose. Betamethasone can push maternal glucose upward for several days, and late preterm treatment also increases neonatal hypoglycemia. In diabetes, that is not a minor detail.

The local GDM protocol gives a practical anchor. During betamethasone administration, the total insulin dose may need to rise by about twenty to thirty percent from the day of the injection for three to five days after Celestone, with meticulous glucose monitoring and individualized adjustment. For patients on oral treatment, a temporary oral dose change or short-term long-acting insulin may be needed.

So the order "give steroids" is incomplete in diabetes.

A complete order includes the obstetric indication, the gestational age, why delivery is likely soon, the glucose monitoring plan, the insulin or medication adjustment plan, and notification to neonatology about neonatal hypoglycemia risk.

This is especially important because the patient and fetus experience the tradeoff differently. The mother may experience hyperglycemia. The neonate may experience hypoglycemia after birth. The respiratory benefit may reduce need for respiratory support. Good care means planning for all of it.

Now consider repeat courses.

The local protocol is clear: after thirty four weeks, a repeat rescue course is not given. In the late preterm window, the question is whether a first course is justified. It is not a rescue-course playground.

This connects to a bigger principle. Repeated fetal steroid exposure raises more concern than a single indicated course. Gabbe's developmental origins material emphasizes that glucocorticoids can influence fetal growth, brain development, H P A axis programming, metabolic function, and cardiovascular biology. The point is not to frighten clinicians away from an indicated course. The point is to avoid unnecessary and repeated exposure when the chance of benefit is low.

## Now let us use three cases.

Case one. A singleton pregnancy at thirty four weeks and two days has ruptured membranes and the team expects delivery within the week. No prior steroid course.

No chorioamnionitis. This is within the late preterm logic. Betamethasone can be considered, with neonatal and glucose planning.

Case two. A patient at thirty five weeks and four days has mild contractions that stop, a closed cervix, and reassuring evaluation. Giving betamethasone "just in case" is weak reasoning. The likelihood of delivery in the next week is not established.

Case three. A patient at thirty four weeks and one day has severe preeclampsia and maternal status requires delivery after stabilization. Steroids should not delay indicated delivery. If there is time and the team judges it safe, a dose may be given. But the steroid course is not the reason to wait.

The resident should hear the structure.

## First, is the patient in the gestational window?

## Second, is delivery likely soon?

## Third, has she already received steroids?

Fourth, is there diabetes or another reason glucose planning must be escalated?

Fifth, would waiting to complete steroids create maternal or fetal danger?

If the answer to the last question is yes, stop. Maternal and fetal safety outrank steroid completion.

**The resident reflex is this: in late preterm care, do not ask only whether betamethasone can reduce respiratory morbidity. Ask what it costs, who was studied, whether birth is really imminent, and how glucose will be managed.**

Late preterm betamethasone is useful when targeted. It becomes poor medicine when it becomes routine reassurance for clinician anxiety.